

CodeAlpha Internship Program

Task 2 — Machine Learning / Deep Learning

# EMOTION RECOGNITION FROM SPEECH

*A Deep Learning Approach Using the RAVDESS Dataset*

## Project Report

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## Abstract

Speech carries not only linguistic content but also rich emotional information that shapes human communication. This project, undertaken as Task 2 of the CodeAlpha internship program, addresses the problem of Speech Emotion Recognition (SER) by building and comparing multiple deep learning approaches on the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS). The dataset consists of 1,440 speech recordings produced by 24 professional actors across eight emotion categories: angry, calm, disgust, fearful, happy, neutral, sad, and surprised. An actor-independent split was used, with actors 01–16 for training, 17–20 for validation, and 21–24 for testing, to ensure that reported performance reflects generalization to unseen speakers rather than memorization of speaker-specific characteristics.

Three principal modelling approaches were implemented and evaluated. The first two approaches used 40-dimensional Mel-Frequency Cepstral Coefficients (MFCCs) extracted at a 16,000 Hz sampling rate as input to a Bidirectional Long Short-Term Memory (BiLSTM) network and a hybrid CNN-LSTM network, respectively. The third approach fine-tuned a pretrained self-supervised speech representation model, Wav2Vec2 XLSR Large (facebook/wav2vec2-large-xlsr-53), with its convolutional feature encoder frozen and a classification head trained for the eight emotion classes. An additional Wav2Vec2 Base model was trained as a supporting baseline to examine the effect of pretrained model scale.

On the held-out, actor-independent test set, the BiLSTM model reached 25.83% accuracy, the CNN-LSTM model reached 27.92% accuracy, the Wav2Vec2 Base model reached 63.33% accuracy, and the fine-tuned Wav2Vec2 XLSR Large model achieved the best performance with 78.75% test accuracy, 82.55% weighted precision, 78.75% recall, and a 78.73% weighted F1-score, at a test loss of 1.7026. These results demonstrate that transfer learning from large-scale self-supervised speech representations provides substantially stronger generalization to unseen speakers than models trained from scratch on handcrafted MFCC features. Wav2Vec2 XLSR Large was therefore selected as the final model, and its predictive capability was made accessible through an interactive Gradio-based web interface that accepts an uploaded audio clip and returns the predicted emotion along with confidence scores. The report discusses the complete pipeline, presents a confusion matrix and classification report analysis, and concludes with the practical applications, limitations, and directions for future improvement of the system.

# 1. Introduction

## 1.1 Speech Emotion Recognition

Speech Emotion Recognition (SER) is the task of automatically identifying the emotional state of a speaker from the acoustic properties of their speech, independent of the linguistic content of what is said. Emotions are conveyed through variations in pitch, energy, speaking rate, spectral shape, and voice quality, all of which can be captured and modelled using signal processing and machine learning techniques. SER sits at the intersection of audio signal processing, affective computing, and deep learning, and has become an increasingly active research area as voice-driven technologies have become part of everyday life.

## 1.2 Importance of Emotion-Aware Speech Systems

As voice assistants, call-center automation, and conversational agents become more widespread, the ability of a system to recognize the emotional tone of a user is an important step toward more natural and empathetic human-computer interaction. An emotion-aware system can adapt its responses, escalate distressed interactions to human operators, or personalize its behaviour based on the affective state of the speaker. This motivates continued research into robust, generalizable emotion recognition models that perform reliably across speakers who were not seen during training.

## 1.3 Role of Machine Learning and Deep Learning

Traditional approaches to SER relied on handcrafted acoustic features paired with classical machine learning classifiers such as Support Vector Machines or Gaussian Mixture Models. Deep learning has since enabled models to learn temporal and hierarchical representations directly from feature sequences or raw audio, improving robustness to speaker and recording variation. More recently, large self-supervised speech models pretrained on thousands of hours of unlabelled audio, such as Wav2Vec2, have made it possible to transfer general-purpose speech representations to downstream tasks like emotion recognition with comparatively small amounts of labelled, task-specific data.

## 1.4 Motivation for This Project

This project was undertaken to gain practical, end-to-end experience with the SER pipeline: from raw audio and dataset preparation, through handcrafted feature extraction and classical deep learning architectures, to fine-tuning a modern pretrained transformer-based speech model. Comparing MFCC-based recurrent architectures against a fine-tuned Wav2Vec2 XLSR model on the same actor-independent split provides direct, empirical insight into how much pretrained self-supervised representations improve generalization for a relatively small, acted-speech emotion dataset such as RAVDESS.

## 2. Problem Statement

Given a short speech recording, the goal of this project is to automatically classify the speaker's emotional state into one of eight discrete categories: angry, calm, disgust, fearful, happy, neutral, sad, or surprised. The central challenge is to build a model that generalizes to speakers who were not part of the training data, since a system that only performs well on speakers it has already heard is of limited practical use. The project therefore evaluates candidate models using an actor-independent test split and compares handcrafted-feature recurrent models against a fine-tuned pretrained speech transformer to determine which approach yields the most reliable emotion predictions on unseen speakers.

### 3. Project Objectives

- To prepare and analyze the RAVDESS speech emotion dataset, including its class distribution and actor-based structure.
- To design an actor-independent train/validation/test split that prevents speaker leakage between splits.
- To extract MFCC-based acoustic features and use them to train and evaluate a BiLSTM model for emotion classification.
- To design and evaluate a hybrid CNN-LSTM architecture that combines convolutional feature extraction with recurrent temporal modelling on the same MFCC features.
- To fine-tune a pretrained self-supervised speech representation model (Wav2Vec2 XLSR Large) for the emotion classification task, and to compare it against a smaller Wav2Vec2 Base model as a supporting baseline.
- To evaluate all models using accuracy, precision, recall, F1-score, and confusion matrix analysis on a common, actor-independent test set.
- To identify the best-performing model and deploy it through an interactive Gradio-based prediction interface.

## 4. Scope of the Project

The scope of this project is limited to categorical emotion classification on the RAVDESS speech-only dataset, using audio-only cues (the RAVDESS song and video/facial-expression modalities were not used). The project covers dataset analysis, feature engineering, model design, training, evaluation, and deployment of a demonstration interface. It does not cover multimodal emotion recognition (e.g., combining audio with facial expressions or text), dimensional emotion modelling (e.g., valence-arousal regression), real-time streaming inference, or deployment to a production environment beyond a temporary, locally hosted Gradio demonstration.

## 5. Technologies and Tools

The project was implemented in Google Colab using a Tesla T4 GPU, and relied on the following core technologies:

| Tool / Library              | Purpose in this Project                                                               |
|-----------------------------|---------------------------------------------------------------------------------------|
| Python                      | Primary programming language used throughout the project.                             |
| Google Colab (Tesla T4 GPU) | Cloud-based notebook environment used for GPU-accelerated model training.             |
| TensorFlow / Keras          | Used to build, train, and evaluate the BiLSTM and CNN-LSTM models.                    |
| PyTorch                     | Backend framework for the Hugging Face Wav2Vec2 models.                               |
| Hugging Face Transformers   | Used to load, fine-tune, and evaluate Wav2Vec2 Base and Wav2Vec2 XLSR Large.          |
| Hugging Face Datasets       | Used to construct audio datasets and apply feature extraction pipelines.              |
| Librosa                     | Used for audio loading, resampling, and MFCC feature extraction.                      |
| NumPy                       | Used for numerical array operations on extracted features.                            |
| Pandas                      | Used for dataset bookkeeping, file indexing, and split assignment.                    |
| Scikit-learn                | Used for evaluation metrics: accuracy, precision, recall, F1-score, confusion matrix. |
| Matplotlib / Seaborn        | Used for data visualization, spectrogram plots, and confusion matrix heatmaps.        |
| Gradio                      | Used to build the interactive web-based emotion prediction interface.                 |

## 6. Dataset Description

The project uses the RAVDESS (Ryerson Audio-Visual Database of Emotional Speech and Song) speech subset, a widely used benchmark dataset for speech emotion recognition research.

### 6.1 Dataset Overview

| Attribute                 | Value                                                |
|---------------------------|------------------------------------------------------|
| Total speech audio files  | 1,440                                                |
| Number of actors          | 24 (12 male, 12 female, per standard RAVDESS design) |
| Number of emotion classes | 8                                                    |
| File format               | 16-bit WAV                                           |

### 6.2 Emotion Classes and File Naming

Each RAVDESS filename encodes metadata as a sequence of numeric identifiers separated by hyphens (e.g., modality, vocal channel, emotion, intensity, statement, repetition, and actor number). The third identifier in the filename corresponds to the emotion code, which was mapped to the eight emotion labels used in this project as follows:

| Emotion Code | Emotion Label |
|--------------|---------------|
| 01           | neutral       |
| 02           | calm          |
| 03           | happy         |
| 04           | sad           |
| 05           | angry         |
| 06           | fearful       |
| 07           | disgust       |
| 08           | surprised     |

### 6.3 Class Distribution

The RAVDESS speech set is close to balanced across emotions, with one notable exception: the neutral class contains only 96 files, exactly half of the 192 files available for each of the other seven emotions. This is because the RAVDESS recording protocol includes two intensity levels (normal and strong) for all emotions except neutral, which is recorded only at normal intensity.

| Emotion | Number of Files |
|---------|-----------------|
| angry   | 192             |

| Emotion   | Number of Files |
|-----------|-----------------|
| calm      | 192             |
| disgust   | 192             |
| fearful   | 192             |
| happy     | 192             |
| neutral   | 96              |
| sad       | 192             |
| surprised | 192             |

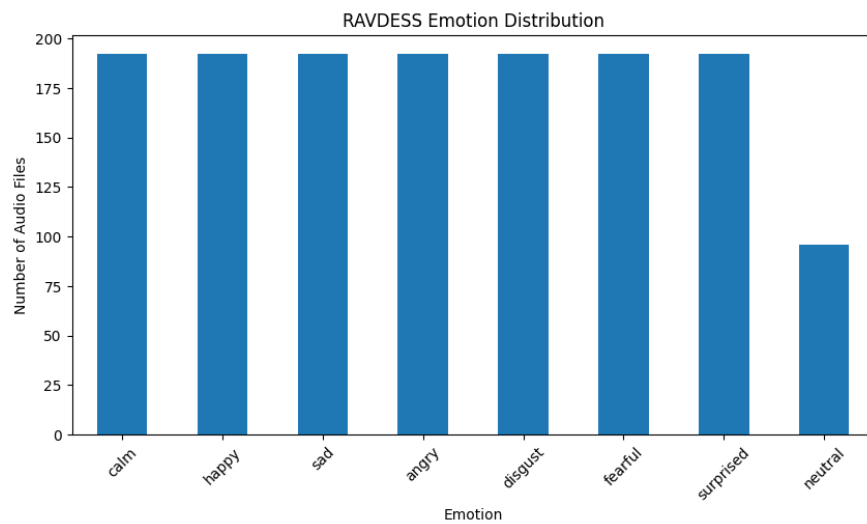


Figure 1. RAVDESS Emotion Distribution

#### 6.4 Actor-Independent Data Split

A speaker-independent (actor-independent) split was used instead of a random file-level split. Under a random split, recordings from the same actor could appear in both the training and test sets, allowing a model to partly rely on speaker-specific vocal characteristics rather than genuinely emotion-discriminative cues, a phenomenon known as speaker leakage. Speaker leakage tends to make test accuracy optimistic and unrepresentative of how the model would perform on a new, unseen speaker in a real deployment. To avoid this, entire actors were assigned exclusively to one split:

- Training set: Actors 01–16
- Validation set: Actors 17–20
- Test set: Actors 21–24

This resulted in 960 training files, 240 validation files, and 240 test files, with no actor appearing in more than one split. Because the split is enforced at the actor level, the reported test performance

reflects each model's ability to generalize its learned emotion cues to speakers it has never encountered, which is a stronger and more realistic evaluation than a random split would provide.

| Split      | Actors | Number of Files |
|------------|--------|-----------------|
| Training   | 01–16  | 960             |
| Validation | 17–20  | 240             |
| Test       | 21–24  | 240             |

#### Emotion Distribution Across Splits

| Emotion   | Train | Validation | Test |
|-----------|-------|------------|------|
| angry     | 128   | 32         | 32   |
| calm      | 128   | 32         | 32   |
| disgust   | 128   | 32         | 32   |
| fearful   | 128   | 32         | 32   |
| happy     | 128   | 32         | 32   |
| neutral   | 64    | 16         | 16   |
| sad       | 128   | 32         | 32   |
| surprised | 128   | 32         | 32   |

## 7. Data Preprocessing

### 7.1 Audio Loading and Resampling

Each audio file was loaded with Librosa and resampled to a uniform sampling rate of 16,000 Hz. Standardizing the sampling rate ensures that all extracted features and the Wav2Vec2 feature extractor operate on audio with a consistent temporal resolution, which is a requirement for both the MFCC-based pipeline and the pretrained Wav2Vec2 models used later in the project.

### 7.2 MFCC Extraction

For the two MFCC-based models, 40 Mel-Frequency Cepstral Coefficients were extracted from each audio signal using Librosa. MFCCs summarize the short-term spectral envelope of speech in a compact, perceptually motivated representation that is commonly used in speech and speaker recognition tasks.

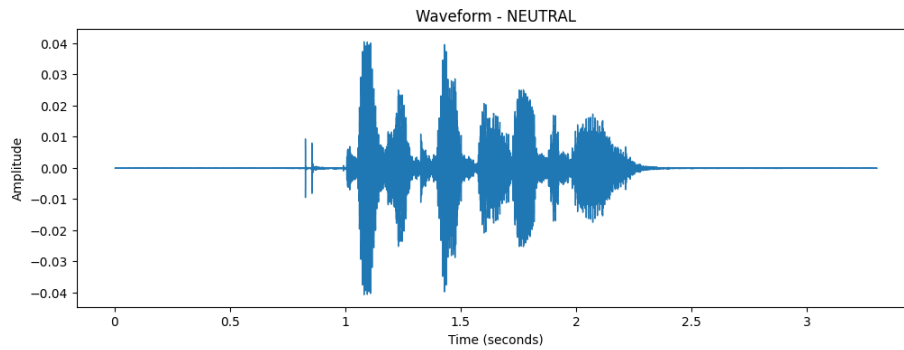


Figure 2. Example Raw Waveform of a RAVDESS Speech Sample

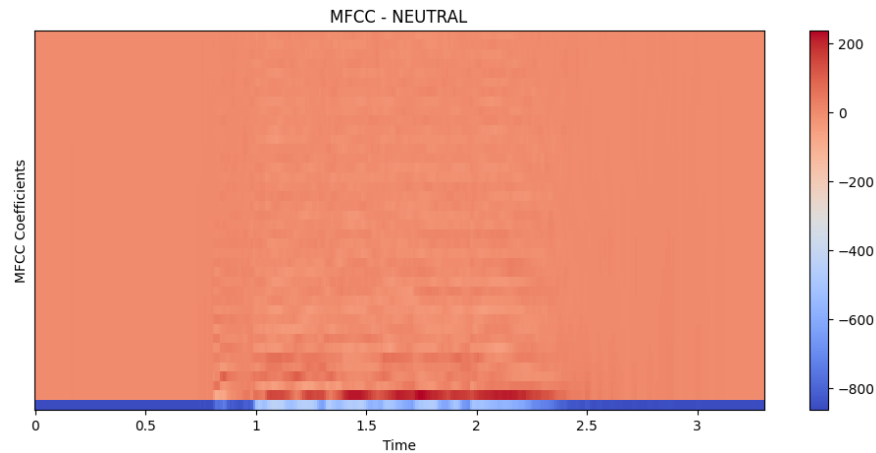


Figure 3. MFCC Feature Representation of the Same Speech Sample

### 7.3 Padding, Truncation, and Fixed-Length Representation

Because speech recordings vary in duration, the extracted MFCC sequences were padded or truncated to a fixed maximum sequence length of 200 frames, producing a consistent MFCC array

of shape (40, 200) — 40 coefficients across 200 time frames — for every sample. This fixed-length representation was required so that batches of samples with different original durations could be processed together by the neural network.

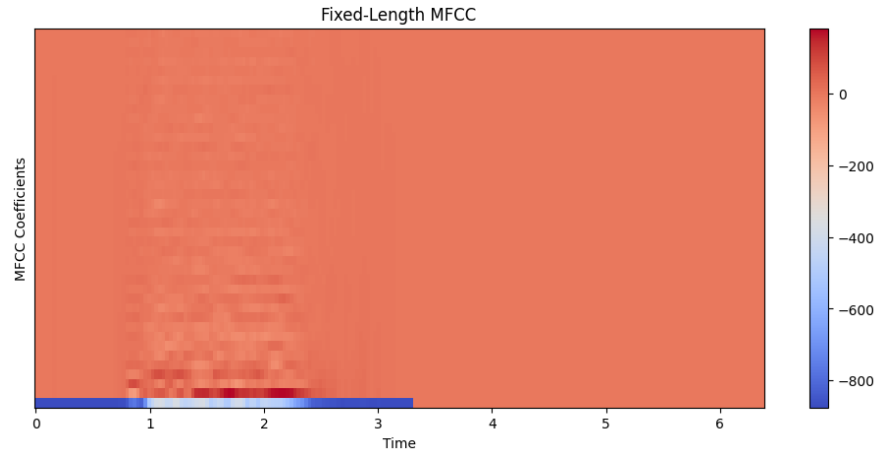


Figure 4. Fixed-Length MFCC Representation After Padding/Truncation

## 7.4 Normalization

Training-set normalization was applied: the mean and standard deviation of the MFCC coefficients were computed exclusively from the training split, and these same statistics were then used to normalize the training, validation, and test sets. Computing normalization statistics only from the training data avoids leaking information from the validation and test sets into the feature scaling process.

## 7.5 Reshaping for the LSTM Input

The MFCC arrays, originally shaped (40, 200) — coefficients by time — were transposed to (200, 40) so that the time dimension came first, matching the sequence-first input convention expected by the recurrent layers (`Input(shape=(200, 40))`) used in both the BiLSTM and CNN-LSTM models.

## 7.6 Label Encoding and Dataset Split

Emotion labels were mapped to integer class indices 0–7 following the order angry, calm, disgust, fearful, happy, neutral, sad, surprised. The MFCC feature arrays and corresponding labels were then partitioned into training, validation, and test arrays using the actor-independent split described in Section 6.4, producing the following array shapes:

| Array   | Shape          |
|---------|----------------|
| X_train | (960, 40, 200) |
| y_train | (960,)         |

| Array  | Shape          |
|--------|----------------|
| X_val  | (240, 40, 200) |
| y_val  | (240,)         |
| X_test | (240, 40, 200) |
| y_test | (240,)         |

## 8. Methodology / Overall System Architecture

The overall system follows a standard supervised speech classification pipeline, implemented in two parallel tracks that share the same dataset and evaluation protocol but differ in their feature representation and model architecture.

### 8.1 End-to-End Pipeline

The complete processing pipeline can be summarized as follows:

- Audio Input — a raw WAV recording from the RAVDESS dataset.
- Preprocessing — resampling to 16,000 Hz and, for the MFCC track, extraction of 40-dimensional MFCCs with padding/truncation to 200 frames and training-set normalization.
- Feature Extraction / Representation — either handcrafted MFCC features (BiLSTM and CNN-LSTM tracks) or learned representations from the pretrained Wav2Vec2 feature extractor and encoder (Wav2Vec2 Base and XLSR Large tracks).
- Deep Learning Model — a BiLSTM, a CNN-LSTM, or a fine-tuned Wav2Vec2 classification head, depending on the track.
- Emotion Classification — a softmax output over the eight emotion classes.
- Prediction / Visualization — the predicted emotion label and confidence score, presented either as evaluation metrics during experimentation or through the Gradio interface during live inference.

### 8.2 Two Modelling Tracks

Track 1 (MFCC-based): raw audio is converted into fixed-length MFCC sequences, which are then modelled with a BiLSTM network (Model 1) and, separately, a CNN-LSTM hybrid network (Model 2). Both models are trained from randomly initialized weights.

Track 2 (Pretrained transfer learning): raw audio is passed through the Wav2Vec2 feature extractor and a pretrained Wav2Vec2 encoder (Base or XLSR Large), whose convolutional feature encoder is frozen. A lightweight classification head on top of the pooled representation is fine-tuned end-to-end for the eight-class emotion task (Model 3, and the supporting Wav2Vec2 Base experiment).

## 9. Model 1 — BiLSTM with MFCC

### 9.1 Why MFCCs Were Used

MFCCs were chosen as the input representation for the first two models because they are a compact, well-established acoustic feature that captures the spectral envelope of speech in a way that approximates human auditory perception. They substantially reduce the dimensionality of the raw waveform while retaining information relevant to timbre and prosody, both of which are known to carry emotional cues.

### 9.2 Bidirectional LSTM

A Long Short-Term Memory (LSTM) network is a recurrent neural network variant designed to model sequential data while mitigating the vanishing-gradient problems of simple recurrent networks. A Bidirectional LSTM (BiLSTM) processes the input sequence in both the forward and backward time directions and concatenates the resulting hidden states, allowing the model to use both past and future context at every time step of the MFCC sequence.

### 9.3 Architecture

| Layer              | Configuration / Output Shape                  |
|--------------------|-----------------------------------------------|
| Input              | (200, 40)                                     |
| Bidirectional LSTM | 128 units, return_sequences=True → (200, 256) |
| Dropout            | —                                             |
| Bidirectional LSTM | → (128)                                       |
| Dropout            | —                                             |
| Dense              | 64 units, ReLU                                |
| Dropout            | —                                             |
| Dense (output)     | 8 units, Softmax                              |

Total trainable parameters: 346,184.

### 9.4 Training Procedure

The model was compiled with the Adam optimizer and sparse categorical cross-entropy loss, and trained for up to 50 epochs. Two callbacks were used: early stopping (monitoring validation loss with a patience of 8 epochs and restoring the best weights) and model checkpointing (saving the model with the highest validation accuracy). During training, training accuracy increased steadily — reaching roughly 68% by the fourteenth epoch — while validation accuracy plateaued in the 25–30% range and validation loss began to increase after the first few epochs, indicating that the model was overfitting the relatively small training set of 960 MFCC sequences.

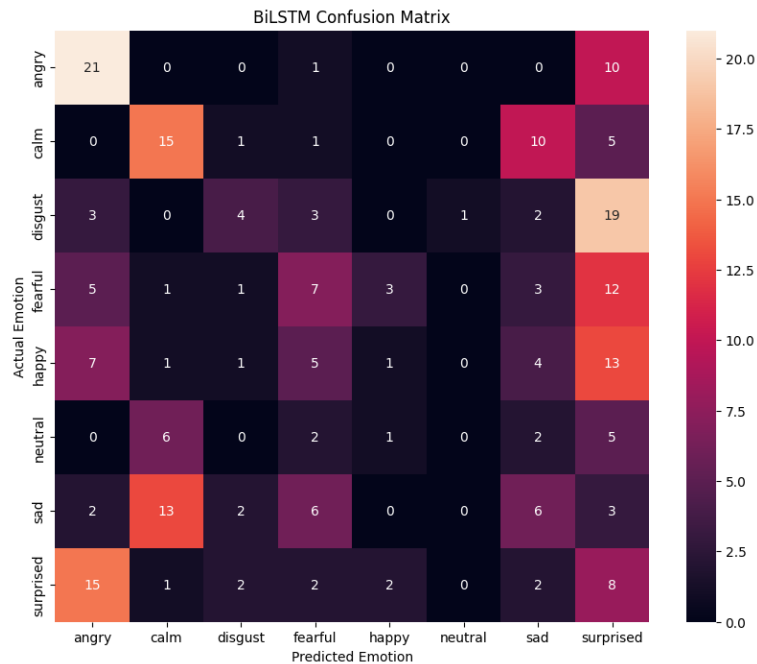
## 9.5 Evaluation and Results

On the held-out, actor-independent test set, the BiLSTM model achieved:

| Metric        | Value  |
|---------------|--------|
| Test Loss     | 1.8068 |
| Test Accuracy | 25.83% |

### Classification Report

| Class              | Precision | Recall | F1-Score | Support |
|--------------------|-----------|--------|----------|---------|
| angry              | 0.3962    | 0.6562 | 0.4941   | 32      |
| calm               | 0.4054    | 0.4688 | 0.4348   | 32      |
| disgust            | 0.3636    | 0.1250 | 0.1860   | 32      |
| fearful            | 0.2593    | 0.2188 | 0.2373   | 32      |
| happy              | 0.1429    | 0.0312 | 0.0513   | 32      |
| neutral            | 0.0000    | 0.0000 | 0.0000   | 16      |
| sad                | 0.2069    | 0.1875 | 0.1967   | 32      |
| surprised          | 0.1067    | 0.2500 | 0.1495   | 32      |
| accuracy (overall) | —         | —      | 0.2583   | 240     |
| macro avg          | 0.2351    | 0.2422 | 0.2187   | 240     |
| weighted avg       | 0.2508    | 0.2583 | 0.2333   | 240     |



*Figure 5. BiLSTM Confusion Matrix*

## 9.6 Discussion: Strengths and Weaknesses

The BiLSTM model was able to learn some emotion-discriminative patterns for angry and calm speech, which achieved the highest recall values among the eight classes. However, its overall test accuracy of 25.83% — only about twice the random-guess baseline of 12.5% for an eight-class problem — indicates that the model struggled to generalize the patterns it learned from 960 training examples to unseen actors. The neutral class was not predicted correctly at all (precision, recall, and F1-score of 0.0000), and disgust, fearful, happy, and sad were also recognized poorly. This behaviour is consistent with the overfitting observed during training: with a comparatively small number of training sequences and no pretrained acoustic knowledge, a BiLSTM trained from scratch on MFCCs has limited capacity to separate emotion classes that share overlapping prosodic characteristics.

## 10. Model 2 — CNN-LSTM with MFCC

### 10.1 CNN Feature Extraction

One-dimensional convolutional layers (Conv1D) were introduced ahead of the recurrent layers to learn local, translation-invariant patterns across the MFCC time axis before temporal modelling. Convolutional filters can act as learned local feature detectors, potentially capturing short-duration spectral patterns (such as transients or short prosodic events) that a purely recurrent model must instead learn to accumulate step by step.

### 10.2 LSTM Temporal Modelling

After the convolutional and pooling stages reduce the temporal resolution and extract local feature maps, two stacked LSTM layers model the longer-range temporal dependencies across the remaining time steps, followed by dense layers that map the final recurrent state to the eight emotion classes.

### 10.3 Architecture

| Layer                                                | Configuration / Output Shape                     |
|------------------------------------------------------|--------------------------------------------------|
| Input                                                | (200, 40)                                        |
| Conv1D                                               | 64 filters, kernel 5, ReLU, same padding         |
| BatchNormalization + MaxPooling1D(2) + Dropout(0.25) | → (100, 64)                                      |
| Conv1D                                               | 128 filters, kernel 3, ReLU, same padding        |
| BatchNormalization + MaxPooling1D(2) + Dropout(0.30) | → (50, 128)                                      |
| LSTM                                                 | 128 units, return_sequences=True + Dropout(0.30) |
| LSTM                                                 | 64 units + Dropout(0.30)                         |
| Dense                                                | 64 units, ReLU + Dropout(0.30)                   |
| Dense (output)                                       | 8 units, Softmax                                 |

Total parameters: 224,008 (223,624 trainable; 384 non-trainable batch-normalization parameters).

### 10.4 Training Procedure

The model was compiled with the Adam optimizer (learning rate 0.001) and sparse categorical cross-entropy loss, and trained for up to 60 epochs with a batch size of 32. Three callbacks were used: early stopping (patience 10, restoring best weights), model checkpointing on best validation accuracy, and a learning-rate reduction schedule (ReduceLROnPlateau, halving the learning rate after 4 epochs without improvement in validation loss, down to a minimum of 1e-6). Before

training, the freshly initialized model was evaluated on the test set as a sanity check and scored 13.75% accuracy — close to the 12.5% level expected from random guessing across eight classes — confirming that the pipeline and evaluation code were functioning correctly prior to training.

## 10.5 Evaluation and Results

After training, the CNN-LSTM model achieved the following performance on the actor-independent test set:

| Metric        | Value  |
|---------------|--------|
| Test Loss     | 1.7761 |
| Test Accuracy | 27.92% |

## 10.6 Discussion

The CNN-LSTM model improved slightly over the plain BiLSTM, raising test accuracy from 25.83% to 27.92% and reducing test loss from 1.8068 to 1.7761. The additional convolutional feature-extraction stage, combined with batch normalization and a learning-rate reduction schedule, appears to have provided a modest regularization and representation benefit over the BiLSTM alone. Nevertheless, both MFCC-based models remain far below the accuracy achieved by the pretrained Wav2Vec2 XLSR model discussed next, underscoring the difficulty of learning strong emotion-discriminative representations from a training set of only 960 examples without the benefit of large-scale pretraining.

## 11. Model 3 — Wav2Vec2 XLSR Large

### 11.1 Wav2Vec2 and Self-Supervised Speech Representations

Wav2Vec2 is a self-supervised speech representation model that learns to represent raw audio by solving a contrastive pretraining task on large amounts of unlabelled speech: portions of the latent representation of the audio are masked, and the model is trained to identify the correct quantized representation for the masked segments among a set of distractors. This pretraining objective allows Wav2Vec2 to learn general-purpose acoustic and phonetic representations without requiring any labelled data, which can then be transferred to downstream tasks such as speech recognition or, as in this project, emotion classification.

### 11.2 XLSR: Cross-Lingual Speech Representations

XLSR (Cross-Lingual Speech Representations) is a variant of Wav2Vec2 pretrained on speech audio from many languages rather than a single language. This cross-lingual pretraining exposes the model to a wider range of phonetic and prosodic variation, which can result in more robust and transferable acoustic representations. The specific checkpoint used in this project was [facebook/wav2vec2-large-xlsr-53](https://huggingface.co/facebook/wav2vec2-large-xlsr-53), a large architecture pretrained on 53 languages.

### 11.3 Transfer Learning and Fine-Tuning

Rather than training a speech model from scratch, this project performs transfer learning: the pretrained XLSR weights are used as a fixed or partially fixed feature extractor, and only a task-specific classification head (plus, where unfrozen, the transformer encoder layers) is updated during training on the RAVDESS emotion labels. This approach allows the model to reuse acoustic knowledge learned from a very large amount of unlabelled multilingual speech, which is particularly valuable given the comparatively small size of the labelled RAVDESS training set (960 examples).

### 11.4 Architecture and Classification Head

The Wav2Vec2ForSequenceClassification architecture was used, which attaches a pooling and linear classification head on top of the Wav2Vec2 encoder outputs. The convolutional feature encoder — the earliest, lowest-level component of the network — was frozen using `freeze_feature_encoder()`, while the remaining encoder and the newly initialized classification head (projector and classifier layers) were fine-tuned. The classification head was configured for 8 output classes matching the RAVDESS emotion labels, with `problem_type` set to `single_label_classification`.

### 11.5 Training Configuration

| Setting                            | Value                                        |
|------------------------------------|----------------------------------------------|
| Base checkpoint                    | facebook/wav2vec2-large-xlsr-53              |
| Learning rate                      | 1e-4                                         |
| Per-device train / eval batch size | 2 / 2                                        |
| Gradient accumulation steps        | 4 (effective batch size 8)                   |
| Number of epochs                   | 15                                           |
| Warm-up steps                      | 180                                          |
| Mixed precision (fp16)             | Enabled                                      |
| Model selection metric             | Weighted F1-score (best checkpoint retained) |
| Hardware                           | Google Colab, NVIDIA Tesla T4 GPU            |

Training ran for 1,800 optimizer steps over 15 epochs, reaching a final training loss of 0.6896 after a total training time of approximately 3,818 seconds (about 63.6 minutes) on the Tesla T4 GPU.

## 11.6 Evaluation and Final Results

| Metric                    | Value  |
|---------------------------|--------|
| Test Loss                 | 1.7026 |
| Test Accuracy             | 78.75% |
| Test Precision (weighted) | 82.55% |
| Test Recall (weighted)    | 78.75% |
| Test F1-Score (weighted)  | 78.73% |

### Classification Report

| Class              | Precision | Recall | F1-Score | Support |
|--------------------|-----------|--------|----------|---------|
| angry              | 0.8000    | 0.8750 | 0.8358   | 32      |
| calm               | 0.9167    | 0.6875 | 0.7857   | 32      |
| disgust            | 0.9286    | 0.8125 | 0.8667   | 32      |
| fearful            | 0.9600    | 0.7500 | 0.8421   | 32      |
| happy              | 0.8889    | 0.5000 | 0.6400   | 32      |
| neutral            | 0.5172    | 0.9375 | 0.6667   | 16      |
| sad                | 0.7568    | 0.8750 | 0.8116   | 32      |
| surprised          | 0.6818    | 0.9375 | 0.7895   | 32      |
| accuracy (overall) | —         | —      | 0.7875   | 240     |

| Class        | Precision | Recall | F1-Score | Support |
|--------------|-----------|--------|----------|---------|
| macro avg    | 0.8062    | 0.7969 | 0.7798   | 240     |
| weighted avg | 0.8255    | 0.7875 | 0.7873   | 240     |

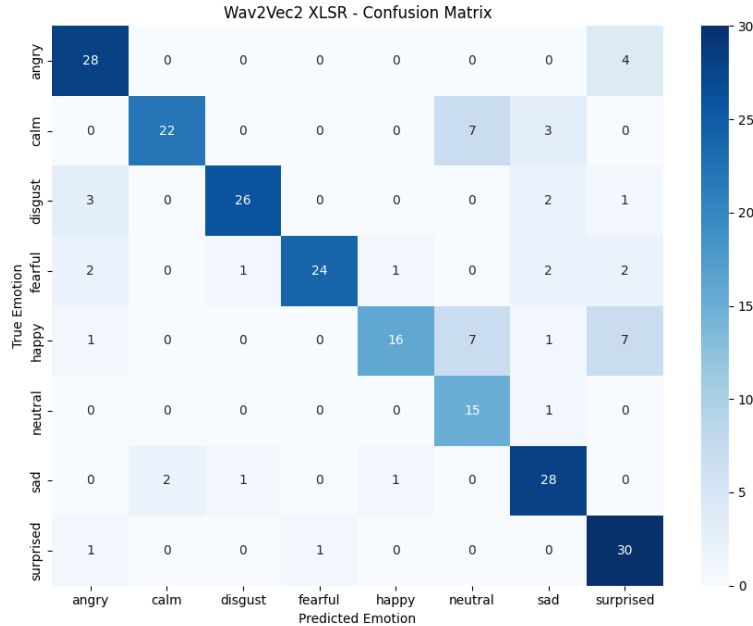


Figure 6. Wav2Vec2 XLSR Large — Final Model Confusion Matrix

### 11.7 Why XLSR Outperformed the MFCC-Based Approaches

The Wav2Vec2 XLSR Large model outperformed both MFCC-based models by a wide margin, more than doubling the test accuracy of the CNN-LSTM model (27.92% → 78.75%). This gap is best explained by the difference in what each model must learn from only 960 labelled training examples. The BiLSTM and CNN-LSTM models must learn emotion-discriminative acoustic patterns entirely from this small training set, using generic MFCC features that were not designed specifically for emotion recognition. The XLSR model, in contrast, begins from representations already shaped by exposure to a very large amount of multilingual speech during self-supervised pretraining, so fine-tuning only needs to adapt and recombine these already informative representations for the emotion classification task, rather than learn acoustic structure from scratch. This transfer-learning advantage is consistent with the broader finding in the speech-processing literature that large pretrained self-supervised models tend to generalize better than models trained from scratch when labelled data is limited.

## 12. Additional Baseline Experiment — Wav2Vec2 Base

As a supporting experiment, and to examine the effect of pretrained model scale, a smaller Wav2Vec2 Base model (facebook/wav2vec2-base) was fine-tuned using the same RAVDESS actor-independent split and a comparable fine-tuning setup: feature encoder frozen, classification head fine-tuned for 8 emotion classes, learning rate 1e-5, effective batch size 8 (batch size 4 with gradient accumulation 2), 10 training epochs, and mixed-precision training on the Tesla T4 GPU. Training ran for 1,200 steps and reached a final training loss of 2.3928 after approximately 819.7 seconds (about 13.7 minutes).

On the actor-independent test set, the Wav2Vec2 Base model achieved:

| Metric        | Value  |
|---------------|--------|
| Test Loss     | 1.1569 |
| Test Accuracy | 63.33% |
| Test F1-Score | 62.94% |

This result is clearly labelled as an additional, supporting experiment rather than one of the three primary models compared in this report. It nonetheless provides a useful data point: the Wav2Vec2 Base model already substantially outperforms both MFCC-based models trained from scratch, confirming that pretrained self-supervised speech representations — even from a comparatively small, monolingual checkpoint — transfer well to the emotion recognition task. Scaling up to the larger, cross-lingually pretrained XLSR checkpoint provided a further, sizeable improvement (63.33% → 78.75% test accuracy), suggesting that both pretraining scale and pretraining diversity (cross-lingual data, in the case of XLSR) contributed to the final model's stronger performance.

### 13. Model Comparison

The table below summarizes the performance of all evaluated models on the common, actor-independent RAVDESS test set.

| Model                             | Feature / Representation                                 | Test Accuracy | Other Metrics                                           | Key Observation                                                                                                           |
|-----------------------------------|----------------------------------------------------------|---------------|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| BiLSTM                            | MFCC (40 coeff., 200 frames)                             | 25.83%        | Loss 1.8068                                             | Trained from scratch; overfit the small training set; neutral class never predicted.                                      |
| CNN-LSTM                          | MFCC (40 coeff., 200 frames)                             | 27.92%        | Loss 1.7761                                             | Slight improvement over BiLSTM via convolutional feature extraction and LR scheduling; still close to low-accuracy range. |
| Wav2Vec2 Base (baseline)          | Pretrained self-supervised (monolingual)                 | 63.33%        | F1 62.94%, Loss 1.1569                                  | Large accuracy jump from transfer learning; supporting experiment, not a primary model.                                   |
| Wav2Vec2 XLSR Large (final model) | Pretrained self-supervised (cross-lingual, 53 languages) | 78.75%        | Precision 82.55%, Recall 78.75%, F1 78.73%, Loss 1.7026 | Best-performing model; selected as the final model for deployment.                                                        |

## 14. Evaluation Metrics

Model performance was assessed using standard classification metrics, defined below in terms of true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN).

### Accuracy

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

The proportion of all predictions that were correct. It is intuitive but can be misleading on imbalanced datasets.

### Precision

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Of all instances predicted as a given class, the proportion that were actually that class. High precision means few false alarms for that class.

### Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Of all instances that actually belong to a given class, the proportion that the model correctly identified. High recall means few missed instances of that class.

### F1-Score

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

The harmonic mean of precision and recall, providing a single balanced measure when both false positives and false negatives matter.

### Confusion Matrix

A confusion matrix is a square table that cross-tabulates the true class (rows) against the predicted class (columns), with each cell showing the number of test samples with that true/predicted combination. Correct predictions fall on the diagonal, while off-diagonal cells reveal specific patterns of confusion between classes.

## 15. Final Model Performance

Wav2Vec2 XLSR Large is the final model selected for this project, on the basis of its clearly superior performance across every evaluation metric compared to the BiLSTM, CNN-LSTM, and Wav2Vec2 Base models.

| Metric                    | Value  |
|---------------------------|--------|
| Test Accuracy             | 78.75% |
| Test Precision (weighted) | 82.55% |
| Test Recall (weighted)    | 78.75% |
| Test F1-Score (weighted)  | 78.73% |
| Test Loss                 | 1.7026 |

An accuracy of 78.75% means that the model correctly classified roughly four out of five speech samples from actors it had never seen during training, across eight emotion classes — a task for which random guessing would be expected to achieve only about 12.5% accuracy. The weighted precision of 82.55% being noticeably higher than recall (78.75%) indicates that, on average, when the model predicted a given emotion it was correct slightly more often than it managed to find all true instances of that emotion; this pattern is visible in the per-class results, where some classes (such as happy) have high precision but comparatively low recall. The weighted F1-score of 78.73%, close to the accuracy figure, confirms that performance is reasonably balanced across the precision/recall trade-off rather than being skewed heavily in one direction. A test loss of 1.7026, while not directly interpretable as a percentage, reflects the average cross-entropy penalty the model incurred on the test set and is consistent with a model that is confident and often correct, but not uniformly certain across all eight classes.

## 16. Confusion Matrix Analysis

The confusion matrix for the final Wav2Vec2 XLSR Large model (Figure 6) shows a strong diagonal, indicating that most test samples were classified correctly. Several patterns are worth highlighting:

- Surprised was recognized perfectly (30 of 30 correctly predicted from the 32 true positives shown in the classification report at 93.75% recall, with the matrix showing 30 correct predictions and 2 misclassifications), and angry, disgust, and sad were also recognized strongly, each with 26 or more of 32 samples correctly classified.
- Happy was the most frequently confused class: of 32 true happy samples, only 16 were correctly predicted, with the remainder split mainly between neutral (7 samples) and surprised (7 samples), suggesting that the acoustic expression of happy in this dataset shares energy and pitch characteristics with both neutral and surprised speech.
- Calm was sometimes confused with neutral (7 of 32 samples) and, to a lesser extent, sad (3 samples) — an intuitive confusion, since calm and neutral speech share low arousal and limited pitch variation.
- Neutral achieved the highest recall of any class (93.75%) but the lowest precision (51.72%), meaning the model correctly identified almost all true neutral samples but also mislabeled some calm and happy samples as neutral, indicating a tendency to over-predict the neutral class.

Overall, the confusion patterns are emotionally coherent: most of the errors occur between emotions that are acoustically or perceptually close (e.g., calm/neutral, happy/surprised), rather than between emotions that would be expected to sound very different (such as angry and calm), which is a reassuring sign that the model has learned genuine emotion-related acoustic structure rather than arbitrary correlations.

## 17. Classification Report Analysis

The per-class classification report for the final model (Section 11.6) shows considerable variation in performance across the eight emotion classes. Fearful achieved the highest precision (96.00%), meaning that when the model predicted fearful it was almost always correct, though its recall (75.00%) shows that a number of true fearful samples were missed. Disgust and calm also show strong precision (92.86% and 91.67% respectively) paired with comparatively lower recall (81.25% and 68.75%), a pattern that recurs across several classes and suggests the model is somewhat conservative — it tends to be confident when it commits to certain emotion labels, at the cost of missing some true instances of those emotions.

At the other end of the scale, neutral shows the reverse pattern: low precision (51.72%) but the highest recall of any class (93.75%), consistent with the confusion-matrix observation that the model over-predicts neutral. Happy has the lowest F1-score among all classes (64.00%), driven by its low recall (50.00%), making it the most difficult emotion for the final model to recognize reliably. The macro-averaged F1-score (77.98%) is close to the weighted F1-score (78.73%), indicating that performance is not dominated by a small number of classes with much larger support; despite neutral having half the support of the other classes (16 versus 32 test samples), overall performance remains broadly consistent across the class-support distribution.

## 18. Prediction Interface / User Interface

To make the final Wav2Vec2 XLSR Large model usable beyond the notebook environment, an interactive prediction interface was built using Gradio and launched as a shareable web application directly from Google Colab.

### 18.1 Audio Upload and Prediction

The interface allows a user to upload or record a speech audio clip. On clicking the predict button, the audio is loaded and resampled to 16,000 Hz, passed through the same Wav2Vec2 XLSR feature extractor and fine-tuned model used during evaluation, and the resulting class probabilities are computed with a softmax layer.

### 18.2 Confidence Score and Top Emotion Probabilities

The interface reports the single most likely emotion along with its confidence score, and separately displays the top three predicted emotions with their associated probabilities (implemented using a Gradio Label component configured to show the top three classes), giving the user a more complete picture of the model's uncertainty rather than only the single highest-probability class.

### 18.3 Final Model Information

The interface also displays a compact model overview panel summarizing the deployed model's key statistics: a test accuracy of 78.75%, an F1-score of 78.73%, support for 8 emotion classes, and the underlying architecture (XLSR), so that a user of the interface can see the model's validated performance alongside each individual prediction.

### 18.4 User Interaction

The interface provides a reset control to clear the current audio input and prediction results, and a system status indicator. As a sanity check, a sample training audio file was passed through the deployed prediction function and correctly classified as neutral with a confidence of 99.98%, confirming that the end-to-end inference pipeline — from raw audio upload to displayed prediction — functioned correctly.

*[Insert Figure 7 Here — Emotion Recognition Prediction Interface: a screenshot of the Gradio web application showing the audio upload panel, the predicted emotion with confidence score, and the model overview panel. No static screenshot of the running interface was included in the supplied notebook file; the interface was launched as a live, temporary public URL (Gradio share link) rather than captured as an image.]*

Figure 7. Emotion Recognition Prediction Interface (placeholder — see note above)

## 19. Results and Discussion

### 19.1 Why BiLSTM Performed Relatively Poorly

The BiLSTM model, trained entirely from randomly initialized weights on only 960 MFCC sequences, achieved a test accuracy of 25.83%. Its training accuracy rose well above its validation accuracy during training, a clear sign of overfitting: the model memorized patterns specific to the training actors rather than learning acoustic cues that generalize to new speakers. With no pretrained acoustic knowledge to draw on, and a training set of this size, the model had limited capacity to separate eight emotion classes purely from MFCC statistics.

### 19.2 Why CNN-LSTM Improved Slightly

Adding convolutional feature-extraction layers ahead of the recurrent layers, along with batch normalization and a learning-rate reduction schedule, gave the CNN-LSTM model a modest edge over the plain BiLSTM (27.92% versus 25.83% test accuracy). The convolutional layers likely helped extract more useful local patterns from the MFCC sequence before the recurrent layers processed longer-range dependencies, and the regularization techniques (dropout, batch normalization, adaptive learning rate) likely reduced overfitting to some degree. However, the improvement remained modest, reinforcing that the underlying limitation was less about the specific recurrent/convolutional architecture and more about the amount of labelled training data available to a model trained from scratch.

### 19.3 Why Pretrained Wav2Vec2 Performed Significantly Better

Both Wav2Vec2 models substantially outperformed the MFCC-based models, with the XLSR Large variant reaching 78.75% test accuracy — roughly three times the accuracy of the CNN-LSTM model. This gap illustrates the central advantage of transfer learning in low-resource settings: because the Wav2Vec2 encoder was pretrained on very large amounts of unlabelled speech before ever seeing RAVDESS, it entered fine-tuning already equipped with general-purpose acoustic representations, leaving the comparatively small labelled RAVDESS training set to be used efficiently for adapting these representations to the emotion task rather than learning basic acoustic structure from zero.

### 19.4 Effect of Learned Speech Representations

The improvement from Wav2Vec2 Base (63.33%) to Wav2Vec2 XLSR Large (78.75%) further shows that the quality and scale of the pretrained representation matters: a larger model pretrained on cross-lingual data captured more transferable acoustic and prosodic structure than the smaller, monolingual Wav2Vec2 Base checkpoint, even though both were fine-tuned on the identical RAVDESS training set with a frozen feature encoder.

### 19.5 Effect of Actor-Independent Testing

Because every model in this project was evaluated using an actor-independent test split, the reported accuracies reflect genuine generalization to unseen speakers rather than any benefit from speaker leakage. This is a more demanding evaluation protocol than a random split, and it is likely that all four models — but especially the smaller MFCC-based models — would report higher, and less representative, accuracy under a random, non-actor-independent split.

## 19.6 Generalization to Unseen Speakers

The final XLSR model's confusion patterns (Section 16) were emotionally coherent rather than arbitrary, which is an encouraging sign that the model generalized genuine acoustic-emotional cues to actors 21–24, none of whom it had seen during training or validation, rather than memorizing speaker-specific characteristics.

## 20. Applications

### Human-Computer Interaction

Enabling voice interfaces and virtual assistants to adapt their tone or response strategy based on the detected emotional state of the user.

### Virtual Assistants

Making conversational agents more context-aware by recognizing frustration, confusion, or satisfaction in a user's voice.

### Call-Center Analysis

Automatically flagging emotionally charged calls (e.g., anger or distress) for review or escalation, and analyzing agent-customer interaction quality.

### E-Learning

Monitoring learner engagement and emotional state during voice-based or video-conferencing educational sessions to help instructors adapt content delivery.

### Well-Being Support Systems

Serving as one signal, among several, within broader well-being applications that flag notable shifts in vocal emotional tone for a user's own awareness — noting that such a system does not provide medical or psychological diagnosis.

### Customer Sentiment / Emotion Analysis

Providing an additional signal for measuring customer satisfaction in voice-based feedback channels.

### Accessibility and Assistive Technologies

Supporting communication tools for individuals who rely on non-verbal cues by surfacing detected vocal emotion alongside transcribed speech.

## 21. Limitations

- RAVDESS is a relatively small dataset (1,440 speech files across 24 actors), which limits the amount of data available for training models from scratch, as reflected in the low accuracy of the BiLSTM and CNN-LSTM models.
- RAVDESS consists of acted, scripted emotional speech performed by professional actors, which may differ systematically from spontaneous, naturally occurring emotional speech encountered in real-world applications.
- The neutral class has only half as many samples (96) as the other seven emotion classes (192 each), an imbalance that is visible in the model's tendency to over-predict neutral and in its comparatively low precision for that class.
- Fine-tuning large pretrained models such as Wav2Vec2 XLSR Large is computationally demanding, requiring GPU acceleration (a Tesla T4 was used) and a training time of over an hour for the final model, which may limit accessibility for users without access to suitable hardware.
- The models were trained and evaluated exclusively on RAVDESS, which was recorded in English by North American actors; performance on other languages, accents, recording conditions, or spontaneous real-world speech was not evaluated in this project and cannot be assumed.

## 22. Future Improvements

- Incorporating larger and more diverse emotional speech datasets, potentially combining RAVDESS with other corpora, to increase training data volume and speaker/language diversity.
- Evaluating performance on real-world, spontaneous speech rather than acted, scripted recordings, to better estimate real-world deployment performance.
- Applying data augmentation techniques (e.g., pitch shifting, time stretching, noise injection, or SpecAugment-style masking) to improve robustness and effectively increase training data diversity.
- Conducting a more systematic hyperparameter optimization (learning rate, batch size, number of epochs, dropout rates) for all four models, rather than relying on a single configuration per model.
- Exploring alternative fine-tuning strategies for the Wav2Vec2 models, such as partially or fully unfreezing the encoder, or using parameter-efficient fine-tuning methods.
- Building ensemble models that combine predictions from the MFCC-based and Wav2Vec2-based models to potentially capture complementary error patterns.
- Extending the approach to multilingual emotion recognition, leveraging XLSR's cross-lingual pretraining more directly by fine-tuning and evaluating on emotional speech datasets in multiple languages.
- Deploying the final model as a persistent web or mobile application (rather than a temporary Gradio share link), with a proper hosting backend for real-world accessibility.

## 23. Conclusion

This project implemented and compared four models for speech emotion recognition on the RAVDESS dataset using a rigorous, actor-independent evaluation protocol: a BiLSTM and a CNN-LSTM model trained from scratch on MFCC features, and a Wav2Vec2 Base model and a Wav2Vec2 XLSR Large model fine-tuned from pretrained self-supervised speech representations. The results showed a clear and consistent pattern: models built on pretrained speech representations substantially outperformed models trained from scratch on handcrafted MFCC features, and the larger, cross-lingually pretrained XLSR checkpoint outperformed the smaller Wav2Vec2 Base checkpoint.

The BiLSTM model reached 25.83% test accuracy, the CNN-LSTM model reached 27.92%, the Wav2Vec2 Base model reached 63.33%, and Wav2Vec2 XLSR Large achieved the best overall performance with 78.75% test accuracy, 82.55% weighted precision, 78.75% recall, and a 78.73% weighted F1-score. Wav2Vec2 XLSR Large was therefore selected as the final model for this project, and its confusion matrix and classification report show that its errors are concentrated among acoustically and perceptually similar emotion pairs (such as calm/neutral and happy/surprised) rather than arbitrary misclassifications, indicating that it learned genuine, generalizable emotion-related acoustic structure. The final model was deployed through an interactive Gradio-based interface, demonstrating a complete, functioning pipeline from raw speech audio to predicted emotion with confidence scores.

## 24. References

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## 25. Appendix

This appendix presents selected implementation snippets referenced in the main report. The full source code is contained in the accompanying project notebook (CodeAlpha\_Emotion\_Pred.ipynb) and is not reproduced here in full.

### A.1 MFCC Extraction Function

```
SAMPLE_RATE = 16000
N_MFCC = 40
MAX_LEN = 200

def extract_mfcc(file_path):
    audio, sr = librosa.load(file_path, sr=SAMPLE_RATE)
    mfcc = librosa.feature.mfcc(y=audio, sr=sr, n_mfcc=N_MFCC)
    if mfcc.shape[1] < MAX_LEN:
        pad = MAX_LEN - mfcc.shape[1]
        mfcc = np.pad(mfcc, ((0, 0), (0, pad)))
    else:
        mfcc = mfcc[:, :MAX_LEN]
    return mfcc
```

### A.2 BiLSTM Model Definition

```
bilstm_model = Sequential([
    Input(shape=(200, 40)),
    Bidirectional(LSTM(128, return_sequences=True)),
    Dropout(0.3),
    Bidirectional(LSTM(64)),
    Dropout(0.3),
    Dense(64, activation="relu"),
    Dropout(0.3),
    Dense(8, activation="softmax")
])
```

### A.3 Wav2Vec2 XLSR Model Loading and Freezing

```
MODEL_NAME_XLSR = "facebook/wav2vec2-large-xlsr-53"

wav2vec_xlsr = Wav2Vec2ForSequenceClassification.from_pretrained(
    MODEL_NAME_XLSR,
    num_labels=8,
    label2id=label2id,
    id2label=id2label,
    problem_type="single_label_classification"
)

wav2vec_xlsr.freeze_feature_encoder()
```

### A.4 Training Arguments (Wav2Vec2 XLSR Large)

```
xlsr_training_args = TrainingArguments(
    output_dir="./wav2vec2-xlsr-emotion",
    eval_strategy="epoch",
    save_strategy="epoch",
    learning_rate=1e-4,
    per_device_train_batch_size=2,
    per_device_eval_batch_size=2,
    gradient_accumulation_steps=4,
    num_train_epochs=15,
    warmup_steps=180,
```

```
logging_steps=25,  
load_best_model_at_end=True,  
metric_for_best_model="f1",  
greater_is_better=True,  
save_total_limit=2,  
report_to="none",  
fp16=True  
)
```